Communication Network Design Lab

(course of Prof. Massimo Tornatore)

Introduction

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PhD candidate

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Lab instructors and communication

Instructors

- Mëmëdhe Ibrahimi, Assistant Professor
  - memdhe.ibrahimi@polimi.it
  - Office: building 20, third floor
- Qiaolun Zhang, PhD candidate
  - qiaolun.zhang@polimi.it
  - Office: building 20, third floor

Email communications

- Include «CND LAB – [description]» in email subject
- Make sure to email your communication to both instructors
- Email us for in-person appointment

Lab material and communication

- Ensure to follow «*CND LAB announcements*» via WeBeep
  - We provide instructions on how and where to download relevant material, including software, slides, and task descriptions
- Material discussed in the class is shared via WeBeep
- Software can be downloaded via this OneDrive repository
  - Link: 02 - LAB 0 (instructions to students)
- Tasks are communicated during each lab session
  - Tasks can be done in groups of up to 3 students
  - Use template provided via WeBeep
    - N.B. Make sure to follow the template and answer all questions

Objectives

1. Apply to practice the theoretical concepts covered during the CND course lectures

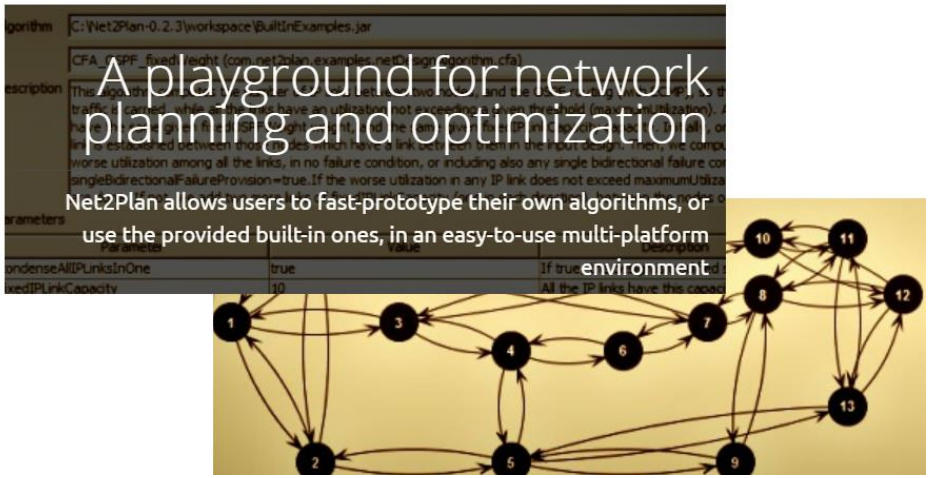
2. Learn to use a network design tool (*Net2Plan*) and python-based libraries for network planning and optimization

3. Learn how to evaluate network performance

Software Used During the Lab

- Net2Plan is a Java-based tool for creating, planning and simulating telecom network instances
  - Publicly-available (open-source)
  - Allows to integrate user-defined algorithms

- Instructions to download Net2Plan and other required components were communicated via email



n2p
net2plan
Net2Plan
The open-source network planner

http://www.net2plan.com/index.php

Useful tips on using Net2Plan

- Net2Plan user manual
  http://www.net2plan.com/documentation/current/help/usersGuide.pdf

- **J**ava **O**ptimization **M**odeler (JOM): Java library targeted to solve optimization problems http://www.net2plan.com/jom/
  - JOM syntax: http://www.net2plan.com/jom/jomSyntax.php
  - Github link: https://github.com/girtel/JOM

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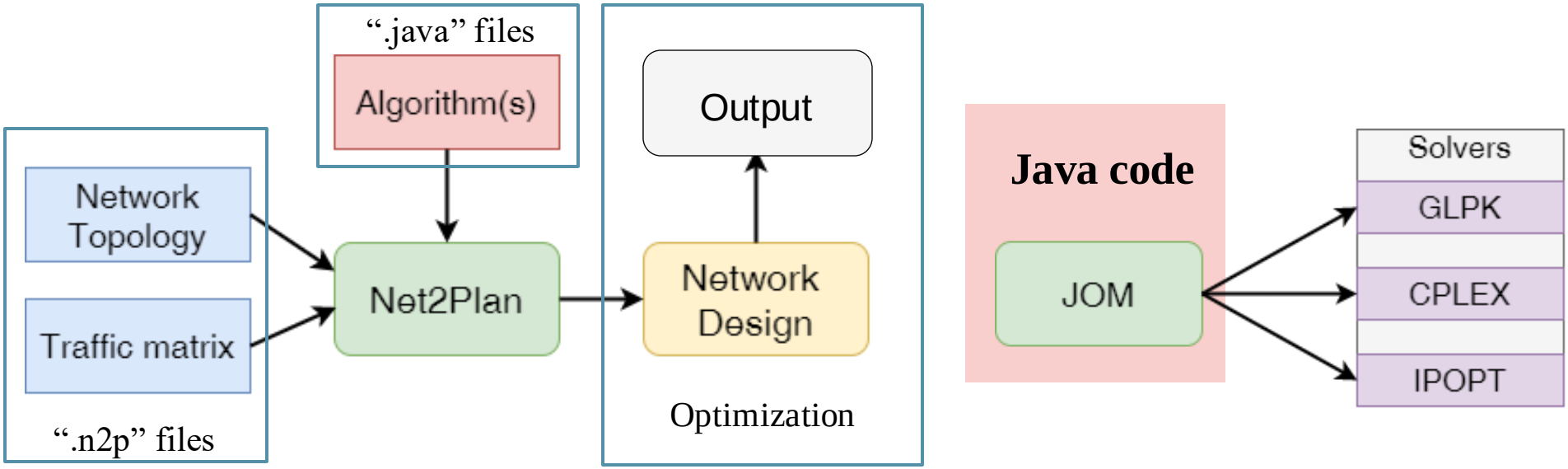

Useful tips on using Net2Plan

- Other useful links to look at
  - http://www.net2plan.com/jom/releases/jom-0.3.2/index.html
  - http://www.net2plan.com/documentation/current/javadoc/examples/
  - http://www.net2plan.com/documentation/current/javadoc/api/index.html

Schedule and Program

- Lab 1 (*29/10/2024*)
  - Introduction to Net2Plan (short demo)
  - Flow Formulation (FF): Splittable and unsplittable flows
  - Task #1: Splittable FF vs. Unsplittable FF
  - Route Formulation (RF) without wavelength continuity
  - RF vs. FF

Prerequisite: Flow formulation and Route Formulation with wavelength continuity

- Lab 2 (0*5/11/2024*)
  - Discussion on Task #1
  - RF with wavelength continuity
  - Task #2: RF *With Wavelength Continuity* vs. *Without Wavelength Continuity*
  - Elastic Optical Networks
    - Routing and Spectrum Assignment (RSA) Problem
  - Heuristic approaches
  - Exercise: RSA problem *unprotected* vs. *1+1 protection*

Prerequisite: Route Formulation without wavelength continuity

Schedule and Program

- Lab 3 (*08/11/2024*)
  - Discussion on Task #2
  - Advanced Topics:
    - IPoWDM Network

Prerequisite:
1+1 protection

- Lab 4 (*15/11/2024*)
  - Python and NetworkX for Network Design
  - Exercise: RSA Problem with ILP using Python
  - If there is time, we can also see other tools for Network Design (e.g., GNPy)
    - https://gnpy.readthedocs.io/en/master/

NetworkX
Network Analysis in Python

Schedule and Program

- Lab 5 (*19/11/2024*)
  - CND projects in collaboration with Nokia
    - Michael Recalcati, Dr. Oleg Karandin, and Olzhas Shaikenov
  - Each group is followed and advised by a senior member of the BONSAI research group or NOKIA
  - Deadline to '*choose the project*' will be announced on the day of the project presentations
  - Estimated project presentation will be in the *end-of-February* to *beginning-of-March*

- NOKIA visit (to be decided)

Evaluation / Bonus Points

- **Task has to be submitted by the set deadline**
  - First task deadline: next lab session 05/11/2024
  - Task ‘weight’ will be announced for each task, e.g., a task may have a max value of 1 point

- Submit **PDF file** (report and discuss results) and **“java” code** via e-mail with subject “***LAB_CND_task_#***”
  - Send email to: qiaolun.zhang@polimi.it AND memedhe.ibrahimi@polimi.it
  - You may use the template for task submission
    - Link: Task submission template